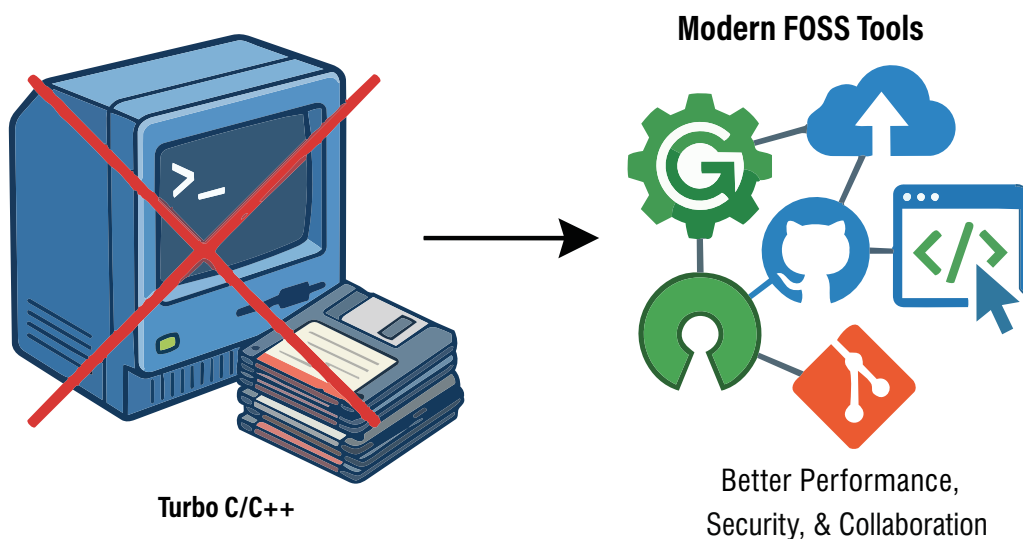


Why It's Time to Replace Turbo C/C++ with Modern FOSS Tools

Thousands of schools and colleges continue to teach C and C++ using Turbo C/C++, a compiler that was last updated more than three decades ago. This outdated software can easily be replaced with free and open source tools that work equally well on Windows and Linux.



Walk into the computer lab of many schools, colleges and coaching institutes, and you will be greeted by a familiar sight: a blue screen with yellow text, running inside a black window. This is Turbo C++, running inside the DOSBox emulator, because modern computers can no longer run it directly. Many of us learnt our first programs on that blue screen, and there is genuine affection for it. But affection is not a good reason to teach with a tool that the rest of the world retired a generation ago.

This old compiler can do concrete harm to students. A common free and open source (FOSS) replacement can work the same way on Windows and Linux, so that no institute needs to change its operating system to make the switch.

A museum piece in the computer lab

Turbo C appeared in the late 1980s, and Turbo C++ 3.0, the version commonly seen in labs today, was released by Borland in 1991. The development of this 16-bit MS-DOS program stopped long ago, and Borland itself no longer exists as an independent company. Since a modern 64-bit Windows system cannot run 16-bit DOS programs at all, institutes install DOSBox, an emulator of a dead operating system, just to run a discontinued compiler to teach a very living language. The emulation brings its own friction, with awkward full-screen behaviour, clumsy copy-paste and needless crashes. To add to this, most copies of Turbo C++ 3.0 circulating on the internet are of

doubtful licence status. We are, in effect, introducing lakhs of students to programming every year on abandoned software of uncertain legality, when legally free and actively maintained alternatives exist.

The language is not the compiler

Before going further, one clarification will make everything else in this article easier to follow, especially for readers who have taught with Turbo C for decades. C and C++ are languages, defined on paper by an international committee under the ISO. The committee publishes a document, called the standard, which states exactly what the language contains: its keywords, its grammar, and its library functions such as `printf()` and